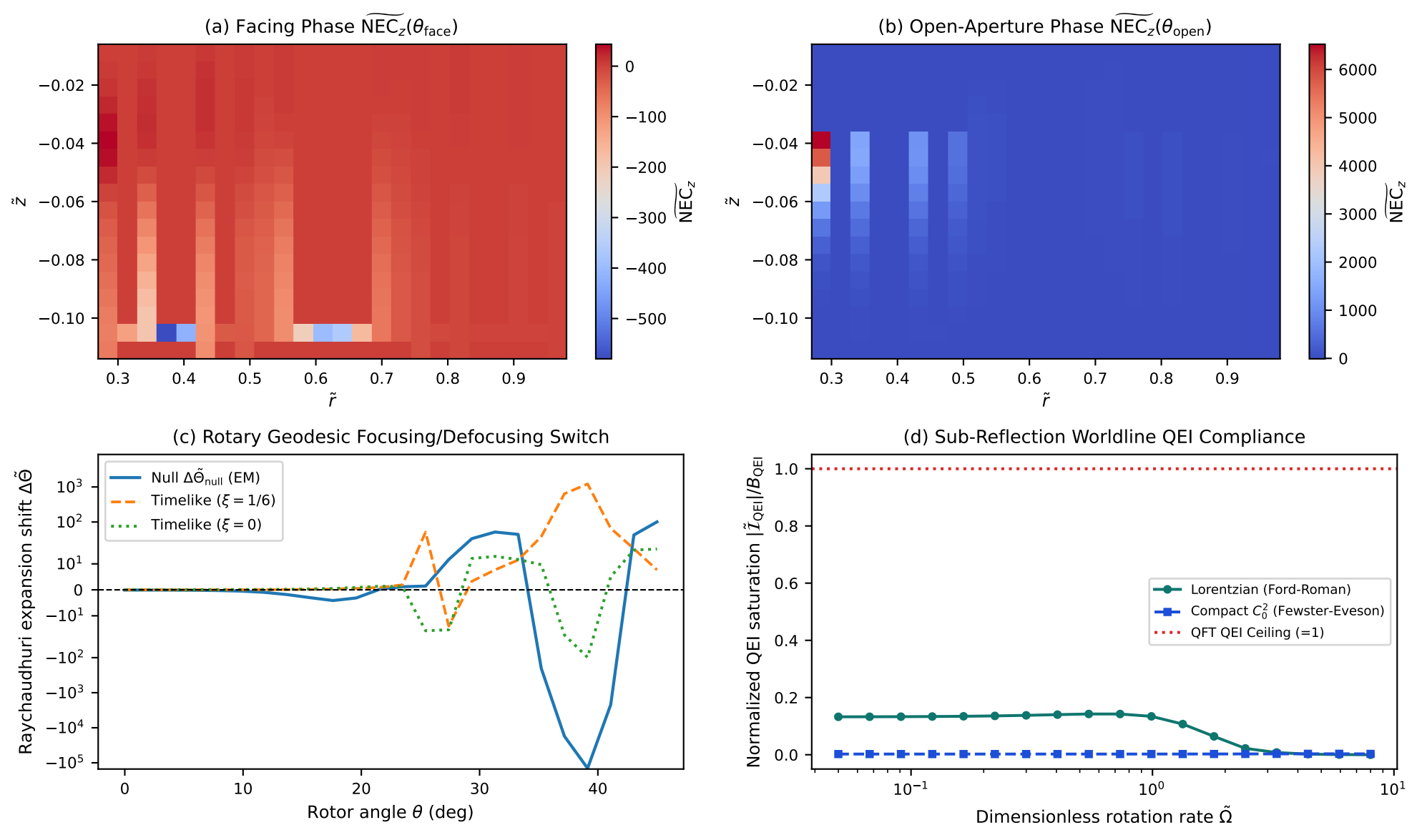
(a) Facing Phase $\widetilde{\mathrm{NEC}_z}(\theta_{\mathrm{face}})$
$\tilde{z}$
$\tilde{r}$
$\widetilde{\mathrm{NEC}_z}$
(b) Open-Aperture Phase $\widetilde{\mathrm{NEC}_z}(\theta_{\mathrm{open}})$
$\tilde{z}$
$\tilde{r}$
$\widetilde{\mathrm{NEC}_z}$
(c) Rotary Geodesic Focusing/Defocusing Switch
Null $\Delta\tilde{\Theta}_{\mathrm{null}}$ (EM)
Timelike ($\xi = 1/6$)
Timelike ($\xi = 0$)
Raychaudhuri expansion shift $\Delta\tilde{\Theta}$
Rotor angle θ (deg)
(d) Sub-Reflection Worldline QEI Compliance
Lorentzian (Ford-Roman)
Compact C_0^2 (Fewster-Eveson)
QFT QEI Ceiling (=1)
Normalized QEI saturation $|\tilde{\mathcal{I}}_{\mathrm{QEI}}|/B_{\mathrm{QEI}}$
Dimensionless rotation rate $\tilde{\Omega}$